

Is artificial intelligence unbiased?

The unequal effect in housing price appraisal

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Introduction: Opening remarks

Artificial intelligence has caused a 50% to 70% decrease in wages—Creating income inequality and threatening millions of jobs (Forbes, 2021).

The rapid growth of AI technology has brought a variety of gains

- Medical, Manufacturing, Logistics areas, etc.
- In the financial area, previous studies have intensively utilized the AI model due to its flexibility. [†]

Introducing AI to housing price appraisal as a prominent instrument

- Specifically, PropTech and FinTech have applied the AI model to manage resources and increase the efficiency of enterprises (Zhang et al., 2021 and Poirier, 2024).
- Based on its flexibility, scholars have actively adopted the AI model in appraising housing prices (Liang et al., 2023; Baur et al., 2023; Chen et al., 2022).

Despite its promising potential, a few negative effects have been reported

- AI can potentially penalize minorities (Desiere and Struyven, 2021; Fiedler et al., 2021).
- This can threaten sustainable urban development and societal equality.

[†] Credit risk appraisal (Fuster et al., 2023), Macroeconomic indicator estimation (Bianchi et al., 2022)

Introduction: What we have done

A (large) discrepancies in housing prices can be the source of

- housing inequality between groups (Larsen and Sommervoll, 2004).
- This can be fueled by high housing prices and the social group's low housing affordability (Zhang et al., 2016).

This study postulates two scenarios: the **emergence of unequal effect** and its **aggravation** in appraising housing prices.

- **The unequal effect**: when the positive difference between assessed housing prices by AI model and HPM is in the well-educated housing group.
 - At the same time, the negative difference is in the less-educated housing group.
- **Aggravation of unequal effect**: it satisfies the following scenario for which
 - the difference between AI model's and HPM's estimated values is statistically larger in well-educated housing groups than that of less-educated housing groups.

Introduction: What we have done

Unveiling the unequal effects embedded in housing price appraisals

- Comparative analysis between AI algorithms and traditional hedonic price models
- The findings reveal that neighborhoods with lower education levels are penalized by AI models
- Whereas, well-educated neighborhoods are favored.

Discrepancies in housing price assessments are exacerbated.

- This occurs in accordance with higher local education levels.
- Implying the tradeoffs of introducing AI algorithms into housing price appraisal
- While taking advantage of flexibility, this loses fairness in housing price assessments.

Data and Methodology: Rationale for the case of Korea

Education has been the primary driver for the advancement of Korea.

- After the Korean War in the early 1950s left the country in ruins, Korea overcame the despair and rapidly grew economically (Oh, 2007).
- The rapid growth of economic development in Korea was possible thanks to the education fever[†] that has been handed down for several hundred years (Jones, 2013; Dittrich and Neuhaus, 2023).
- Since the end of the 1990s, demand for excellent creative manpower has grown, fostering knowledge-intensive cutting-edge technology industries in tandem with changes in the global economy[‡].

[†] Korean society has traditionally long respected the need and desire for learning, which worked as the driving force in making Korean an education stronghold (Korean Education Centre UK, 2015).

[‡] Thus, Korea has focused on developing human resources for a knowledge-based society, requiring well-educated workers and professional industrial manpower (Korean Education Centre UK, 2015).

Data and Methodology: Rationale for the case of Korea

Increasing educational costs in private tutoring

- Education became a way to socially succeed and the fairest way of climbing up the economic ladder in Korean society.
- However, criticisms are mounting that this intensive education fever is exacerbating the polarization of society because it forces parents to excessively invest in children's private tutoring (Korean Education Centre UK, 2015)[†].
- Hence, the increased educational cost is imputed to housing costs, burdening lower-income families to reside in highly educated cities.
- This corroborates the existing literature documenting housing cost barriers related to the education (Berry and Glaseser, 2005; Choi and Green, 2022); thus, better education services and institutions raise neighborhood housing prices (Nguyen-Hoang and Yinger, 2011; Wen et al., 2017).

In this regard, we note that the case study of Korea is proper to scrutinize whether the unequal effect of AI-based housing price appraisal is exacerbated in accordance with educational levels.

[†] Households in Korea with a monthly income of eight million Korean won (5,848.80 USD) or more spent an average of 671,000 won (490.57 USD) per month on their child's private education whereas, a monthly income of two million won (1,462.20 USD) or less spent an average of 136,000 won (99.43 USD) per month for private education services. The average monthly expenditure per student for private education was approximately 431,000 won (315.10 USD) (Statista, 2023).

Data and Methodology: Data

The data periods are aggregated with 2018 and 2019 alongside four metropolitan cities: Busan, Daegu, Daejeon, and Gwangju.

- A total of 17 variables are used in this study including two education variables.

We choose two education variables, 'Higher Degree' and 'Top Univ.' as proxies for the education levels in urban cities.

- **Higher Degree**: Percentage of the number of university graduates divided by the number of people aged 15 years or older
 - It has been conventionally utilized as a proxy for education levels in relation to housing prices (Ahn et al., 2020; Lin et al., 2022; An et al., 2023)
 - A higher degree has served as the requisite credential for occupying qualified job opportunities in Korea[†]; thus, there is a belief that people with a higher degree can afford the greater housing costs based on their higher wages[‡].

[†] The enrollment rate at universities was 52.5 percent in 2000 and has constantly increased to 76.2 percent in 2023 (Statista, 2023).

[‡] University graduates are more likely to settle for higher housing costs due to the better public services, economic potential, and urban infrastructure (Lin et al., 2022)

Data and Methodology: Data

Top Univ.: Number of Seoul National University entrants from high schools within a 5-km radius of properties

- While hundreds of universities and community colleges are in Korea, entrance to the top university is highly competitive because the degree from good universities has a higher potential for a high social standing than the degree from lower-ranked universities[†].
- Our variable, "Top Univ.", can thus represent the extent of education fever or the quality of educational services in the neighborhood.

[†] Also, graduates from higher-ranked universities tend to have higher housing affordability in the long run (Li et al., 2017)

Data and Methodology: Methodology

Housing price appraisal models

Traditional hedonic price model (HPM)

- OLS-based HPM
- Spatial lag regression (to resolve the spatial dependency)

Machine learning (ML) algorithms

- Random forest (RF)
 - XGBoost
- These models have the flexibility to capture the nonlinearity embedded in dataset.
- So, previous studies have widely employed these algorithms to precisely appraise housing prices.
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Data and Methodology: Experiment setting

We divide the whole dataset into the train and test dataset

- Training set with a fraction of 0.7, test set with a fraction of 0.3.
 - We construct three cases: (1) training a model by controlling *Higher Degree*, (2) by controlling *Top. Univ.*, and (3) by omitting two education variables.
 - After training each model, we evaluate the test dataset for appraising housing prices ($\hat{y}$)
 - We calculate the difference between the estimated values of AI model and HPM, i.e., $\hat{y}_{AI} - \hat{y}_{HPM}$.
 - We sort the differences in ascending order based on the education variable
→ and aggregate lower and upper groups into 5, 10, and 20 percentiles.
 - Lastly, we conduct the t -test to discriminate whether the discrepancy is exacerbated on average according to the neighborhood's education level.
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Results and discussion: Regression results

Busan						
Obs. = 53,458	(i)		(ii)		(iii)	
	OLS	Spatial lag	OLS	Spatial lag	OLS	Spatial lag
Property Characteristics						
Size	0.012‡	0.011‡	0.012‡	0.011‡	0.012‡	0.011‡
Floor	0.005‡	0.004‡	0.005‡	0.004‡	0.005‡	0.004‡
Units	$7.7 \times 10^{-5} \ddagger$	$7.1 \times 10^{-5} \ddagger$	$8.1 \times 10^{-5} \ddagger$	$7.4 \times 10^{-5} \ddagger$	$7.6 \times 10^{-5} \ddagger$	$7.0 \times 10^{-5} \ddagger$
Parking	0.159‡	0.134‡	0.166‡	0.139‡	0.160‡	0.134‡
Heating	0.155‡	0.092‡	0.154‡	0.089‡	0.141‡	0.081‡
Year	0.012‡	0.012‡	0.012‡	0.012‡	0.012‡	0.012‡
Environmental Amenities						
Dist. Green	-0.008‡	-0.005‡	-0.008‡	-0.005‡	-0.008‡	-0.005‡
Dist. Water	-0.014‡	-0.012‡	-0.018‡	-0.015‡	-0.018‡	-0.014‡
Local Built Environment						
Bus stop	$-3.3 \times 10^{-4} \ddagger$	$2.7 \times 10^{-4} \ddagger$	-0.001‡	-4.6×10^{-5}	0.001‡	0.001‡
Dist. Subway	-0.075‡	-0.055‡	-0.080‡	-0.057‡	-0.076‡	-0.054‡
Top Univ.	0.006‡	0.004‡	0.006‡	0.004‡		
Local Demographics						
Sex Ratio	-0.007‡	-0.004‡	-0.011‡	-0.007‡	-0.008‡	-0.004‡
Pop. Density	$2.4 \times 10^{-6} \ddagger$	$1.7 \times 10^{-6} \ddagger$	$1.9 \times 10^{-6} \ddagger$	$1.3 \times 10^{-6} \ddagger$	$2.9 \times 10^{-6} \ddagger$	$2.0 \times 10^{-6} \ddagger$
Higher Degree	0.005‡	0.004‡			0.005‡	0.004‡
Seasonality Control						
Spring	0.018‡	0.017‡	0.020‡	0.018‡	0.017‡	0.016‡
Fall	0.074‡	0.064‡	0.078‡	0.066‡	0.075‡	0.064‡
Winter	0.048‡	0.044‡	0.050‡	0.045‡	0.047‡	0.044‡
Constant	-14.150‡	-17.267‡	-12.870‡	-16.403‡	-14.204‡	-17.437‡
VIF range	1.060–1.828		1.060–1.826		1.048–1.828	
F-statistics	7,250‡	7,200‡	7,510‡	7,450‡	7,600‡	7,540‡
RMSE	0.313	0.291	0.316	0.293	0.315	0.292
Adjusted R-square	0.698	0.696	0.692	0.690	0.695	0.693

Note: ‡ and † indicate significance at the levels of 1% and 5%, respectively.

Results and discussion: Regression results

Obs. = 56,606	Daegu					
	(i)		(ii)		(iii)	
	OLS	Spatial lag	OLS	Spatial lag	OLS	Spatial lag
Property Characteristics						
Size	0.011‡	0.011‡	0.011‡	0.011‡	0.011‡	0.011‡
Floor	0.004‡	0.004‡	0.005‡	0.004‡	0.004‡	0.004‡
Units	-3.8×10^{-7}	$4.6 \times 10^{-6} \dagger$	$1.0 \times 10^{-5} \ddagger$	$1.3 \times 10^{-5} \ddagger$	$-5.7 \times 10^{-6} \dagger$	1.8×10^{-6}
Parking	0.068‡	0.050‡	0.087‡	0.060‡	0.074‡	0.051‡
Heating	-0.016‡	-0.008‡	0.004	0.008‡	-0.041‡	-0.023‡
Year	0.022‡	0.020‡	0.020‡	0.018‡	0.023‡	0.020‡
Environmental Amenities						
Dist. Green	-0.012‡	-0.011‡	-0.021‡	-0.017‡	0.001	-0.002*
Dist. Water	0.019‡	0.009‡	0.013‡	0.003‡	0.030‡	0.015‡
Local Built Environment						
Bus stop	-0.001‡	-7.2×10^{-5}	-0.001‡	$5.8 \times 10^{-4} \ddagger$	0.001‡	0.002‡
Dist. Subway	-0.043‡	-0.036‡	-0.069‡	-0.053‡	-0.054‡	-0.043‡
Top Univ.	0.009‡	0.006‡	0.011‡	0.008‡		
Local Demographics						
Sex Ratio	-0.003‡	-0.004‡	-0.005‡	-0.005‡	-0.005‡	-0.005‡
Pop. Density	$1.1 \times 10^{-6} \ddagger$	$1.2 \times 10^{-6} \ddagger$	$2.2 \times 10^{-6} \ddagger$	$1.9 \times 10^{-6} \ddagger$	$3.4 \times 10^{-6} \ddagger$	$2.8 \times 10^{-6} \ddagger$
Higher Degree	0.012‡	0.009‡			0.015‡	0.011‡
Seasonality Control						
Spring	-0.021‡	-0.014‡	-0.016‡	-0.009‡	-0.022‡	-0.014‡
Fall	0.026‡	0.030‡	0.036‡	0.038‡	0.025‡	0.030‡
Winter	-0.002	0.004	0.007‡	0.010‡	-0.003	0.004
Constant	-35.737‡	-33.511‡	-30.252‡	-29.349‡	-36.038‡	-33.381‡
VIF range	1.150–1.689		1.144–1.546		1.128–1.578	
F-statistics	12,000‡	11,300‡	11,200‡	10,800‡	11,400‡	10,900‡
RMSE	0.249	0.234	0.262	0.241	0.260	0.240
Adjusted R-square	0.783	0.772	0.760	0.753	0.763	0.756

Note: ‡, †, and * indicate significance at the levels of 1%, 5%, and 10%, respectively.

Results and discussion: Regression results

Daejeon						
Obs. = 24,350	(i)		(ii)		(iii)	
	OLS	Spatial lag	OLS	Spatial lag	OLS	Spatial lag
Property Characteristics						
Size	0.013‡	0.012‡	0.013‡	0.012‡	0.013‡	0.012‡
Floor	0.003‡	0.004‡	0.003‡	0.004‡	0.002‡	0.003‡
Units	$1.1 \times 10^{-4} \ddagger$	$1.5 \times 10^{-4} \ddagger$	$1.2 \times 10^{-4} \ddagger$	$1.5 \times 10^{-4} \ddagger$	$1.1 \times 10^{-4} \ddagger$	$1.4 \times 10^{-4} \ddagger$
Parking	0.149‡	0.142‡	0.152‡	0.143‡	0.131‡	0.127‡
Heating	0.184‡	0.103‡	0.189‡	0.102‡	0.199‡	0.110‡
Year	0.019‡	0.018‡	0.019‡	0.018‡	0.017‡	0.016‡
Environmental Amenities						
Dist. Green	0.023‡	0.021‡	0.020‡	0.019‡	0.043‡	0.036‡
Dist. Water	-0.012‡	0.014‡	-0.012‡	0.015‡	-0.012‡	0.016‡
Local Built Environment						
Bus stop	-0.003‡	0.003‡	-0.004‡	$2.7 \times 10^{-3} \ddagger$	8.3×10^{-7}	0.005‡
Dist. Subway	-0.045‡	-0.050‡	-0.049‡	-0.052‡	-0.067‡	-0.068‡
Top Univ.	0.012‡	0.010‡	0.014‡	0.010‡		
Local Demographics						
Sex Ratio	-0.013‡	-0.005‡	-0.013‡	-0.005‡	-0.010‡	-0.003‡
Pop. Density	$-7.6 \times 10^{-6} \ddagger$	$-3.7 \times 10^{-6} \ddagger$	$-8.1 \times 10^{-6} \ddagger$	$-3.7 \times 10^{-6} \ddagger$	$-2.3 \times 10^{-6} \ddagger$	$8.4 \times 10^{-7} \ddagger$
Higher Degree	0.003‡	0.001‡			0.007‡	0.004‡
Seasonality Control						
Spring	-0.054‡	-0.044‡	-0.055‡	-0.044‡	-0.053‡	-0.042‡
Fall	0.051‡	0.049‡	0.051‡	0.049‡	0.052‡	0.050‡
Winter	0.002	0.004	0.003	0.004	0.001	0.003
Constant	-28.916‡	-30.045‡	-28.845‡	-30.054‡	-23.638‡	-25.848‡
VIF range	1.119–2.519		1.101–2.024		1.083–1.654	
F-statistics	4,160‡	3,940‡	4,400‡	4,150‡	4,040‡	3,820‡
RMSE	0.281	0.252	0.281	0.252	0.290	0.259
Adjusted R-square	0.744	0.733	0.743	0.732	0.726	0.715

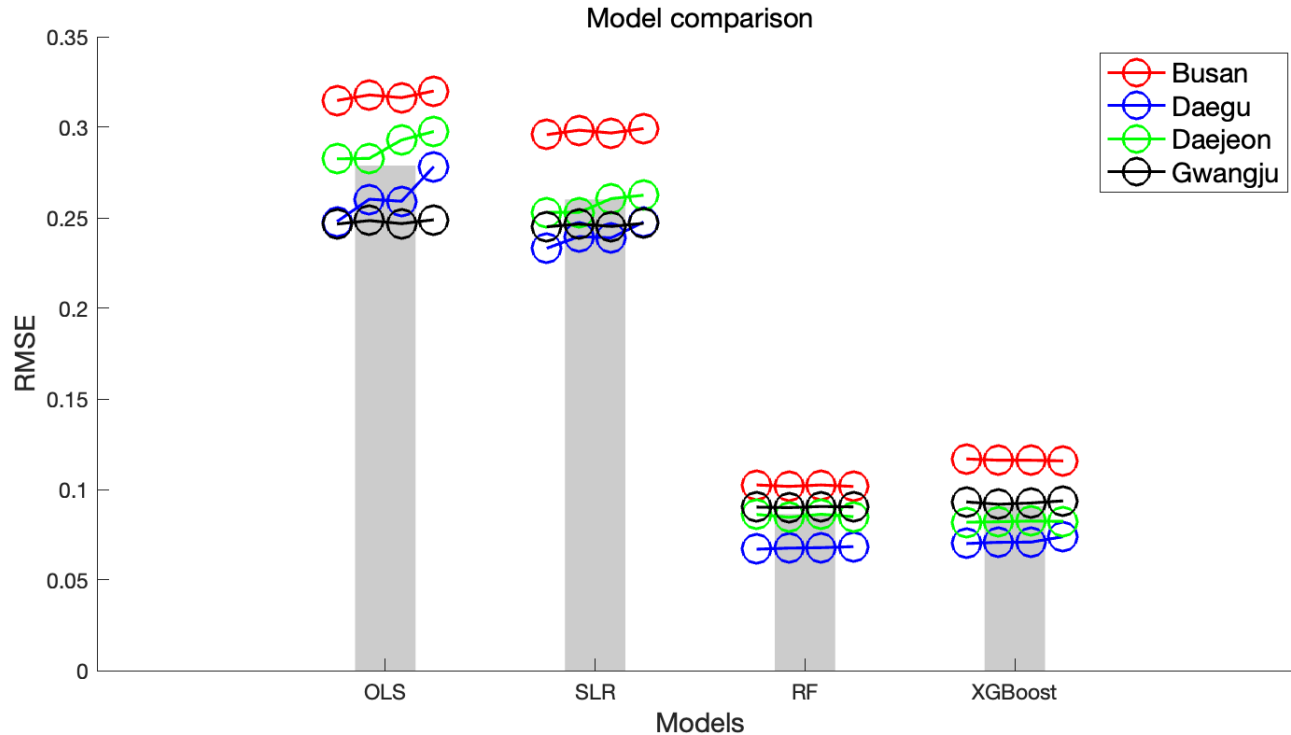
Note: ‡ indicates the significance level of 1%.

Results and discussion: Regression results

Gwangju						
Obs. = 44,305	(i)		(ii)		(iii)	
	OLS	Spatial lag	OLS	Spatial lag	OLS	Spatial lag
Property Characteristics						
Size	0.013‡	0.013‡	0.013‡	0.013‡	0.013‡	0.013‡
Floor	0.002‡	0.002‡	0.002‡	0.002‡	0.002‡	0.002‡
Units	$1.4 \times 10^{-4} \ddagger$	$1.4 \times 10^{-4} \ddagger$	$1.3 \times 10^{-4} \ddagger$	$1.3 \times 10^{-4} \ddagger$	$1.5 \times 10^{-4} \ddagger$	$1.5 \times 10^{-4} \ddagger$
Parking	0.252‡	0.250‡	0.265‡	0.262‡	0.251‡	0.250‡
Heating	0.128‡	0.118‡	0.121‡	0.112‡	0.125‡	0.116‡
Year	0.022‡	0.022‡	0.022‡	0.021‡	0.022‡	0.022‡
Environmental Amenities						
Dist. Green	-0.018‡	-0.013‡	-0.016‡	-0.012‡	-0.017‡	-0.013‡
Dist. Water	-0.033‡	-0.033‡	-0.032‡	-0.032‡	-0.032‡	-0.032‡
Local Built Environment						
Bus stop	0.002‡	0.002‡	0.002‡	$2.2 \times 10^{-3} \ddagger$	$2.2 \times 10^{-3} \ddagger$	0.002‡
Dist. Subway	-0.027‡	-0.029‡	-0.024‡	-0.027‡	-0.027‡	-0.029‡
Top Univ.	0.001‡	0.001‡	0.001‡	0.001‡		
Local Demographics						
Sex Ratio	-0.007‡	-0.007‡	-0.010‡	-0.010‡	-0.007‡	-0.008‡
Pop. Density	$3.2 \times 10^{-6} \ddagger$	$2.8 \times 10^{-6} \ddagger$	$3.0 \times 10^{-6} \ddagger$	$2.6 \times 10^{-6} \ddagger$	$3.3 \times 10^{-6} \ddagger$	$2.9 \times 10^{-6} \ddagger$
Higher Degree	0.005‡	0.004‡			0.005‡	0.004‡
Seasonality Control						
Spring	-0.015‡	-0.015‡	-0.015‡	-0.015‡	-0.015‡	-0.015‡
Fall	0.026‡	0.026‡	0.026‡	0.026‡	0.026‡	0.026‡
Winter	-0.004	-0.005	-0.004	-0.004	-0.004	-0.004
Constant	-33.945‡	-34.093‡	-33.208‡	-33.454‡	-33.928‡	-34.076‡
VIF range	1.051– 2.047		1.037– 2.024		1.085– 2.047	
F-statistics	9,860‡	9,810‡	10,300‡	10,300‡	10,400‡	10,400‡
RMSE	0.245	0.244	0.247	0.245	0.245	0.244
Adjusted R-square	0.791	0.790	0.788	0.788	0.790	0.790

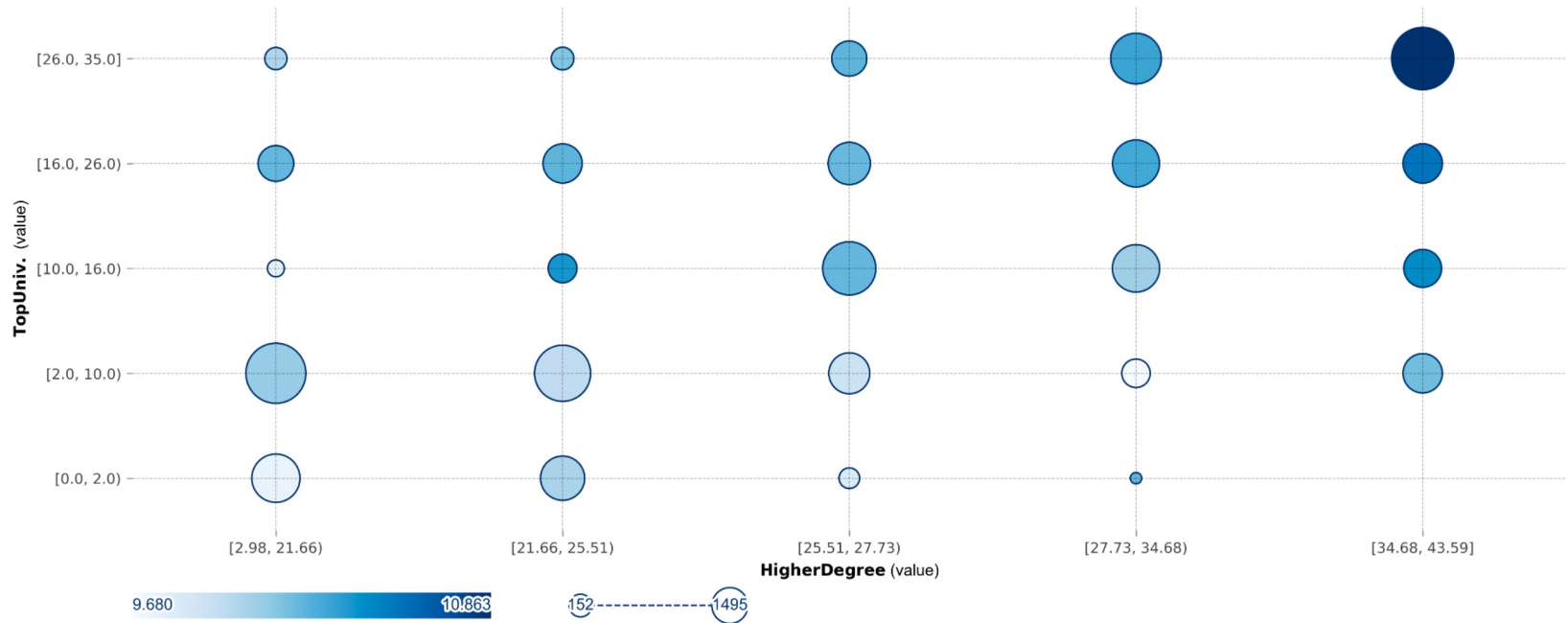
Note: ‡ indicates the significance level of 1%.

Results and discussion: Assessment of housing prices



Each circle denotes the RMSE value corresponding to four cases: (1) Full variables, (2) Omitting 'Higher Degree' from case (1), (3) Omitting 'Top Univ.' from case (1), and (4) Omitting 'Higher Degree' and 'Top Univ.' from case (1). The gray bar represents the average RMSE value of four cases.

Results and discussion: Assessment of housing prices



This figure shows the modality of AI algorithm according to two education variables ('Higher Degree' and 'Top Univ.') in appraising housing prices[†].

The bigger circle indicates the more observations are grouped.

The greater saturation of a circle represents the higher median values of estimated housing prices.

[†] We visualize the case of Daegu when RF is used in the assessment of housing prices because it shows the lowest RMSE value regarding all the cases. We checked that other cases also have the similar results.

Results and discussion: Disparity in appraisal

	Busan					
	5%	95%	10%	90%	20%	80%
Panel A.1: 'Higher Degree' is omitted.						
RF-OLS	0.043	0.183	0.019	0.135	0.005	0.099
<i>t</i> -test	***		***		***	
RF-SLR	0.027	0.178	0.021	0.129	0.007	0.084
<i>t</i> -test	***		***		***	
XGB-OLS	0.043	0.180	0.018	0.128	0.003	0.094
<i>t</i> -test	***		***		***	
XGB-SLR	0.027	0.174	0.019	0.123	0.005	0.080
<i>t</i> -test	***		***		***	
Panel A.2: 'Higher Degree' and 'Top Univ.' are omitted.						
RF-OLS	0.037	0.205	0.012	0.150	-0.001	0.101
<i>t</i> -test	***		***		***	
RF-SLR	0.023	0.193	0.015	0.139	0.003	0.086
<i>t</i> -test	***		***		***	
XGB-OLS	0.043	0.202	0.013	0.145	-0.003	0.097
<i>t</i> -test	***		***		***	
XGB-SLR	0.030	0.189	0.017	0.134	0.001	0.082
<i>t</i> -test	***		***		***	

	Busan					
	5%	95%	10%	90%	20%	80%
Panel B.1: 'Top Univ.' is omitted.						
RF-OLS	-0.088	0.081	-0.079	0.069	-0.041	0.053
<i>t</i> -test	***		***		***	
RF-SLR	-0.062	0.054	-0.052	0.043	-0.022	0.035
<i>t</i> -test	***		***		***	
XGB-OLS	-0.082	0.075	-0.076	0.065	-0.042	0.051
<i>t</i> -test	***		***		***	
XGB-SLR	-0.056	0.048	-0.049	0.039	-0.022	0.033
<i>t</i> -test	***		***		***	
Panel B.2: 'Higher Degree' and 'Top Univ.' are omitted.						
RF-OLS	-0.104	0.095	-0.093	0.078	-0.049	0.059
<i>t</i> -test	***		***		***	
RF-SLR	-0.074	0.064	-0.062	0.049	-0.027	0.039
<i>t</i> -test	***		***		***	
XGB-OLS	-0.097	0.096	-0.089	0.079	-0.049	0.059
<i>t</i> -test	***		***		***	
XGB-SLR	-0.067	0.065	-0.058	0.051	-0.027	0.039
<i>t</i> -test	***		***		***	

Panel A aggregates lower and upper groups based on 'Higher Degree' according to 5-, 10-, and 20-percentiles.

Panel B aggregates lower and upper groups based on 'Top Univ.' according to 5-, 10-, and 20-percentiles.

Each reported value indicates the average of difference between estimated housing prices by AI algorithm and HPM.

For instance, Panel A.1 represents the *t*-test result for lower and upper groups, aggregated by 'Higher Degree' when the housing prices are appraised without 'Higher Degree.'

Results and discussion: Disparity in appraisal

	Daegu					
	5%	95%	10%	90%	20%	80%
Panel A.1: 'Higher Degree' is omitted.						
RF-OLS	-0.044	0.059	-0.081	0.123	-0.082	0.137
<i>t</i> -test	***		***		***	
RF-SLR	-0.025	0.032	-0.057	0.078	-0.058	0.091
<i>t</i> -test	***		***		***	
XGB-OLS	-0.044	0.061	-0.081	0.122	-0.083	0.138
<i>t</i> -test	***		***		***	
XGB-SLR	-0.025	0.034	-0.057	0.077	-0.059	0.092
<i>t</i> -test	***		***		***	
Panel A.2: 'Higher Degree' and 'Top Univ.' are omitted.						
RF-OLS	-0.065	0.118	-0.093	0.138	-0.092	0.179
<i>t</i> -test	***		***		***	
RF-SLR	-0.035	0.064	-0.060	0.080	-0.060	0.109
<i>t</i> -test	***		***		***	
XGB-OLS	-0.065	0.120	-0.095	0.137	-0.093	0.178
<i>t</i> -test	***		***		***	
XGB-SLR	-0.035	0.067	-0.062	0.079	-0.061	0.108
<i>t</i> -test	***		***		***	

	Daegu					
	5%	95%	10%	90%	20%	80%
Panel B.1: 'Top Univ.' is omitted.						
RF-OLS	-0.038	0.014	-0.053	0.080	-0.058	0.105
<i>t</i> -test	***		***		***	
RF-SLR	-0.055	0.006	-0.054	0.050	-0.043	0.071
<i>t</i> -test	***		***		***	
XGB-OLS	-0.040	0.017	-0.054	0.082	-0.059	0.106
<i>t</i> -test	***		***		***	
XGB-SLR	-0.058	0.009	-0.055	0.052	-0.044	0.072
<i>t</i> -test	***		***		***	
Panel B.2: 'Higher Degree' and 'Top Univ.' are omitted.						
RF-OLS	-0.109	0.020	-0.110	0.098	-0.112	0.142
<i>t</i> -test	***		***		***	
RF-SLR	-0.105	0.009	-0.091	0.055	-0.074	0.087
<i>t</i> -test	***		***		***	
XGB-OLS	-0.112	0.023	-0.112	0.099	-0.113	0.143
<i>t</i> -test	***		***		***	
XGB-SLR	-0.108	0.011	-0.092	0.056	-0.075	0.088
<i>t</i> -test	***		***		***	

Results and discussion: Disparity in appraisal

	Daejeon					
	5%	95%	10%	90%	20%	80%
Panel A.1: 'Higher Degree' is omitted.						
RF-OLS	-0.042	0.022	0.013	-0.038	0.001	0.021
<i>t</i> -test	***		***		*	
RF-SLR	-0.038	0.072	-0.016	0.002	0.016	0.013
<i>t</i> -test	***					
XGB-OLS	-0.048	0.019	0.010	-0.040	0.001	0.018
<i>t</i> -test	***		***			
XGB-SLR	-0.043	0.068	-0.020	0.000	0.016	0.009
<i>t</i> -test	***					
Panel A.2: 'Higher Degree' and 'Top Univ.' are omitted.						
RF-OLS	-0.091	0.085	-0.039	0.062	-0.044	0.068
<i>t</i> -test	***		***		***	
RF-SLR	-0.074	0.123	-0.058	0.080	-0.016	0.046
<i>t</i> -test	***		***		***	
XGB-OLS	-0.096	0.085	-0.042	0.062	-0.045	0.067
<i>t</i> -test	***		***		***	
XGB-SLR	-0.078	0.123	-0.061	0.080	-0.017	0.045
<i>t</i> -test	***		***		***	

	Daejeon					
	5%	95%	10%	90%	20%	80%
Panel B.1: 'Top Univ.' is omitted.						
RF-OLS	-0.188	0.134	-0.111	0.056	-0.060	0.084
<i>t</i> -test	***		***		***	
RF-SLR	-0.025	0.005	-0.041	0.042	-0.059	0.065
<i>t</i> -test	*		***		***	
XGB-OLS	-0.182	0.137	-0.107	0.056	-0.058	0.081
<i>t</i> -test	***		***		***	
XGB-SLR	-0.020	0.008	-0.038	0.041	-0.057	0.063
<i>t</i> -test	*		***		***	
Panel B.2: 'Higher Degree' and 'Top Univ.' are omitted.						
RF-OLS	-0.215	0.163	-0.132	0.087	-0.081	0.120
<i>t</i> -test	***		***		***	
RF-SLR	-0.033	0.016	-0.050	0.059	-0.072	0.086
<i>t</i> -test	***		***		***	
XGB-OLS	-0.218	0.164	-0.132	0.086	-0.083	0.117
<i>t</i> -test	***		***		***	
XGB-SLR	-0.036	0.016	-0.051	0.058	-0.073	0.083
<i>t</i> -test	***		***		***	

Results and discussion: Disparity in appraisal

	Gwangju					
	5%	95%	10%	90%	20%	80%
Panel A.1: 'Higher Degree' is omitted.						
RF-OLS	-0.088	0.028	-0.054	0.032	-0.055	0.055
<i>t</i> -test	***		***		***	
RF-SLR	-0.077	0.027	-0.046	0.028	-0.049	0.049
<i>t</i> -test	***		***		***	
XGB-OLS	-0.084	0.030	-0.052	0.032	-0.054	0.056
<i>t</i> -test	***		***		***	
XGB-SLR	-0.073	0.029	-0.044	0.028	-0.048	0.050
<i>t</i> -test	***		***		***	
Panel A.2: 'Higher Degree' and 'Top Univ.' are omitted.						
RF-OLS	-0.091	0.034	-0.059	0.035	-0.059	0.058
<i>t</i> -test	***		***		***	
RF-SLR	-0.080	0.032	-0.051	0.030	-0.052	0.052
<i>t</i> -test	***		***		***	
XGB-OLS	-0.091	0.036	-0.057	0.034	-0.058	0.058
<i>t</i> -test	***		***		***	
XGB-SLR	-0.080	0.034	-0.050	0.030	-0.052	0.052
<i>t</i> -test	***		***		***	

	Gwangju					
	5%	95%	10%	90%	20%	80%
Panel B.1: 'Top Univ.' is omitted.						
RF-OLS	-0.008	-0.030	-0.031	-0.010	-0.019	0.014
<i>t</i> -test	*		**		***	
RF-SLR	-0.004	-0.032	-0.028	-0.012	-0.016	0.014
<i>t</i> -test	**		*		***	
XGB-OLS	-0.007	-0.024	-0.029	-0.008	-0.017	0.014
<i>t</i> -test			**		***	
XGB-SLR	-0.003	-0.026	-0.026	-0.010	-0.014	0.013
<i>t</i> -test	*		*		***	
Panel B.2: 'Higher Degree' and 'Top Univ.' are omitted.						
RF-OLS	-0.014	-0.033	-0.040	-0.008	-0.025	0.022
<i>t</i> -test			***		***	
RF-SLR	-0.010	-0.035	-0.036	-0.010	-0.020	0.021
<i>t</i> -test	**		***		***	
XGB-OLS	-0.010	-0.034	-0.038	-0.009	-0.022	0.020
<i>t</i> -test	**		***		***	
XGB-SLR	-0.006	-0.036	-0.034	-0.011	-0.018	0.019
<i>t</i> -test	**		**		***	

Results and discussion: Relievable inequality

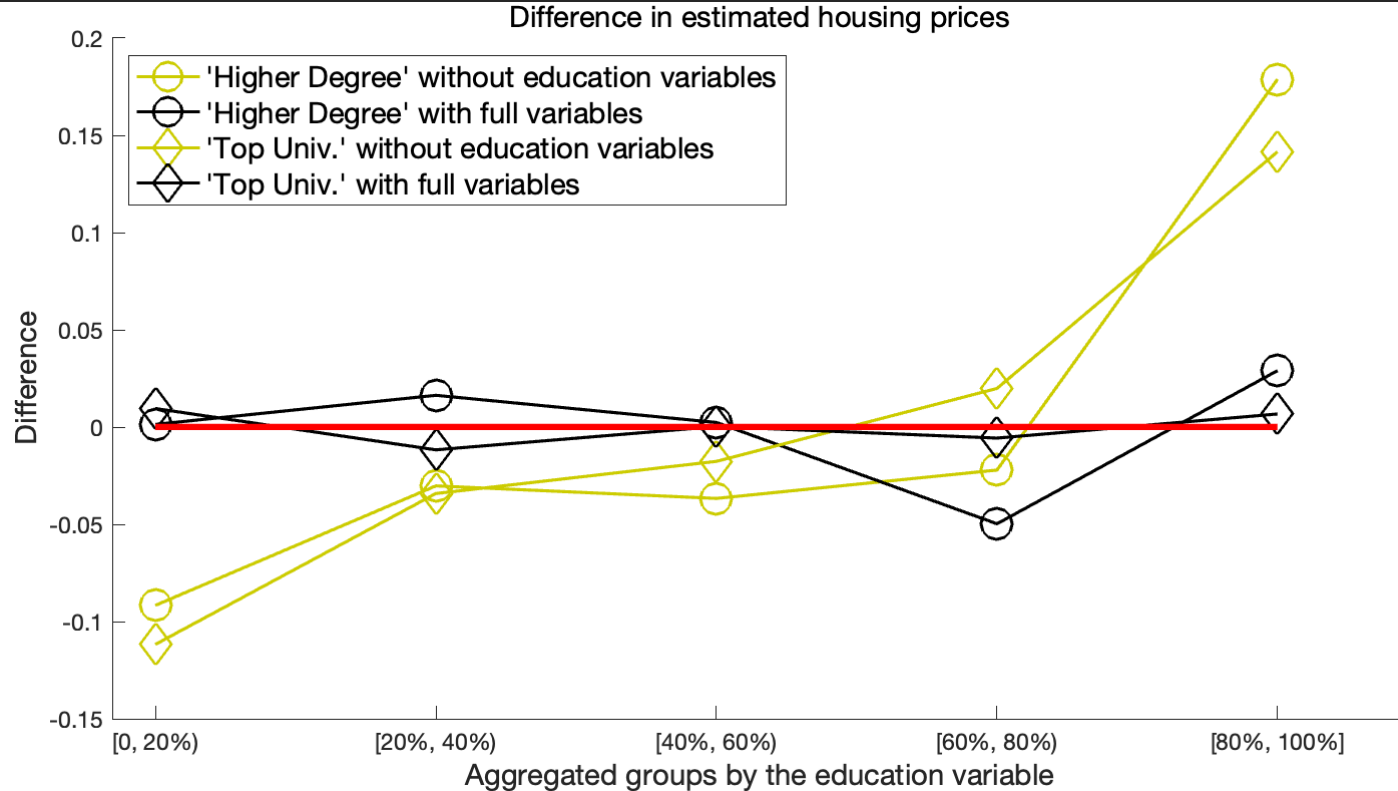
We suggest one alternative way to alleviate this inequality in the assessment of housing prices.

The inequality in housing prices appraisal is incurred when education variables are omitted.

Thus, we further investigate whether the unequal effect can be relieved if two education variables are controlled in the explanatory variable set.

We consider Daegu with RF as the representative case as it shows the most precise accuracy in the assessment of housing prices.

Results and discussion: Relievable inequality



The yellow line indicates the difference between estimated housing prices without two education variables, 'Higher Degree' and 'Top Univ.', and the black line represents the discrepancy between estimated housing prices with full variables including two education variables.

Each circle is the case of aggregating each group based on 'Higher Degree' and the diamond symbol stands for each group aggregated by 'Top Univ.'

Conclusion

- This study investigates the unequal effects of introducing AI algorithms to housing price appraisals.
 - The comparative analysis demonstrates that
 - Housing prices in less-educated neighborhoods are strongly depreciated by AI algorithms.
 - Conversely, housing prices in well-educated neighborhoods are appreciated by AI algorithms with favored valuations.
 - The discrepancy between lower and upper groups is exacerbated according to local educational levels, caused by the flexibility of ML algorithms.
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